EET202

HINES & TRANSFORMERS

MODULE:2

EMF Equation of DC Generator and Types of Excitation

e performance characteristics of DC generators

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INTRODUCTION

rinciple of DC Generator

Law of Electromagnetic Induction

the flux linked with a coil
n EMF will be induced in the coil.

EMF will be proportional to the
nge of flux.

EMF Equation

r pole in Weber (Wb)

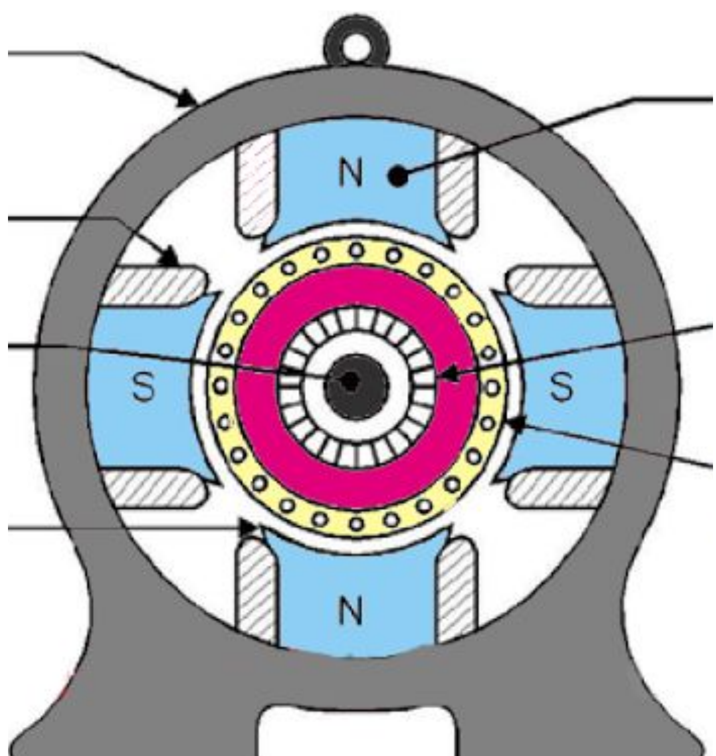
umber of armature conductors

r of poles

er of parallel paths

of armature in rpm

enerated in volts



F Equation (contd..)

ductor in one revolution of armature, dϕ = Pϕ

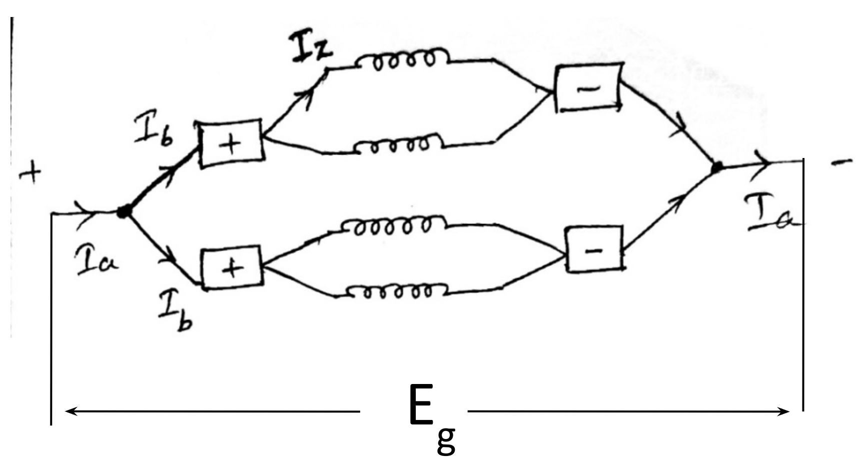
plete one revolution, dt = 60/N sec

one conductor, $\frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{PN\phi}{60}$

rated, E_g = (EMF per conductor) × number of
ductors in one parallel path

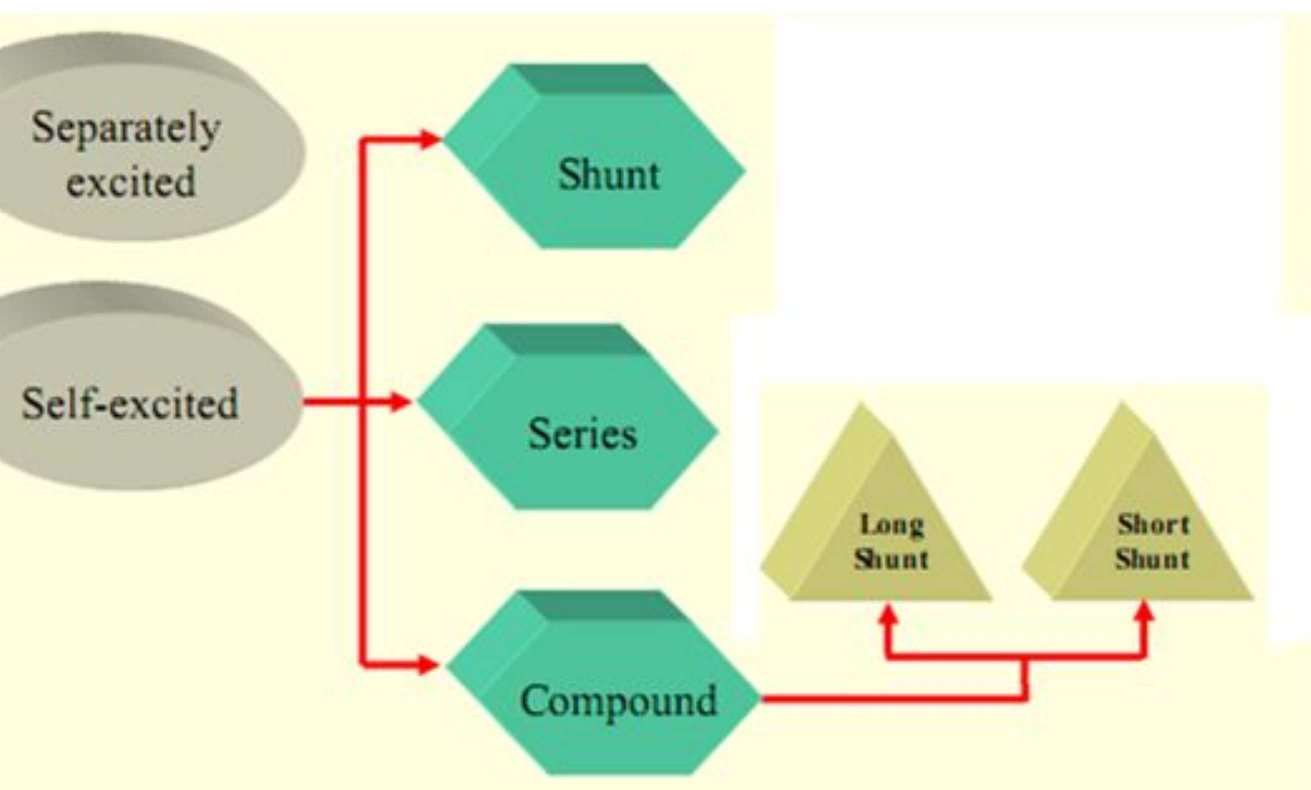
$$= \frac{PN\phi}{60} \times \frac{Z}{A}$$

$$\boxed{\times \frac{P}{A}}$$



es of DC Generators

of Excitation

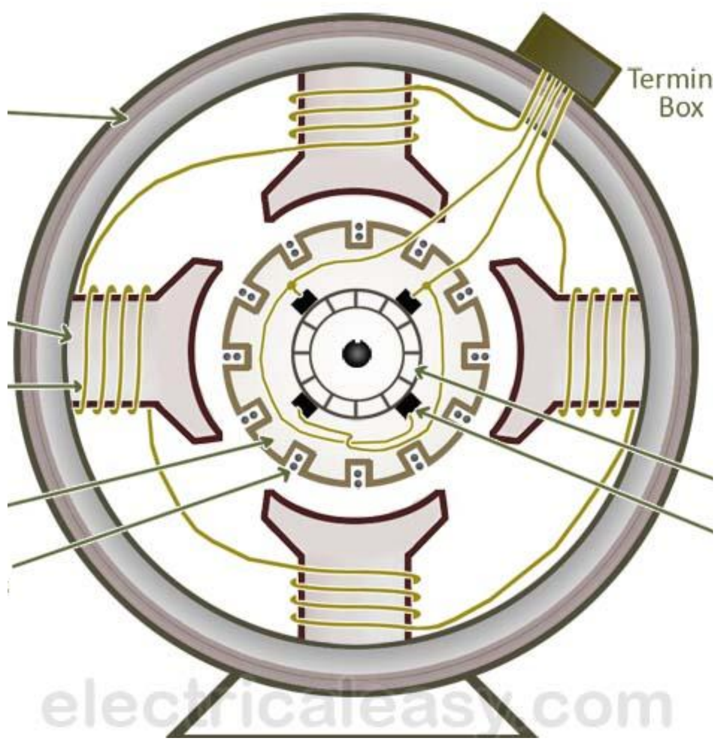


es of DC Generators

action to take place there should be magnetic by field winding.

ited DC Generator – is excited from an ernal dc source

C Generator – If field cited by the current generator itself.



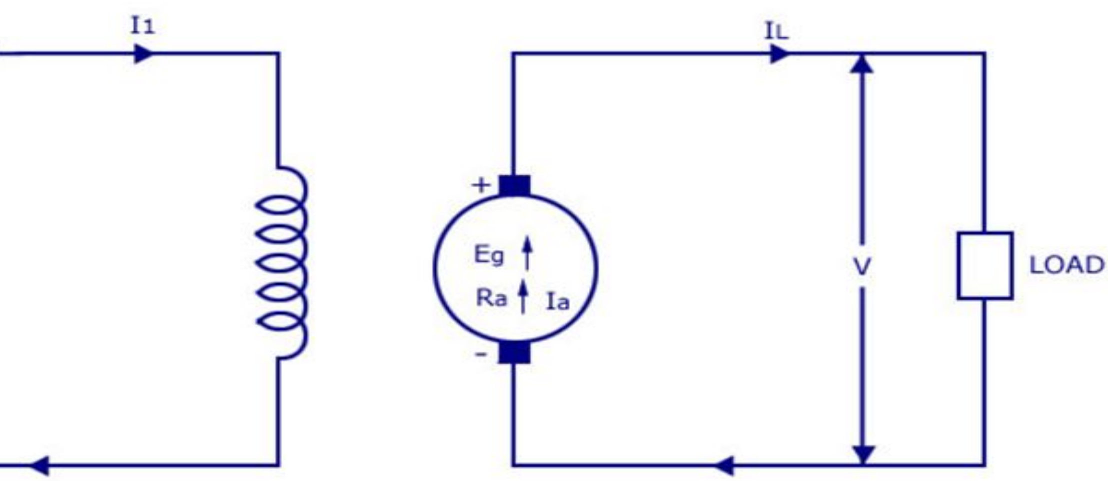
tely Excited DC Generator

arately excited DC generator is ne whose field winding is ed by an independent external rce (like a battery).

agnitude of generated voltage ds upon the speed of rotation of ure and the field current, i.e., r the speed and the field t, higher is the generated e.

ctice, the separately excited DC tors are rarely used

ely Excited DC Generator



$$\text{Armature current, } I_a = I_L$$

$$\text{Terminal voltage, } V_T = E_g - I_a R_a$$

$$\text{Developed electric power} = E_g I_a$$

$$\text{Power delivered to load} = E_g I_a - I_a^2 R_a = V_T I_a = V_T I_L$$

f Excited DC Generator

C generator is the one whose field winding is urrent from the output of the generator itself. ne connection of field winding with the armature, C generators are of three types –

.

erator

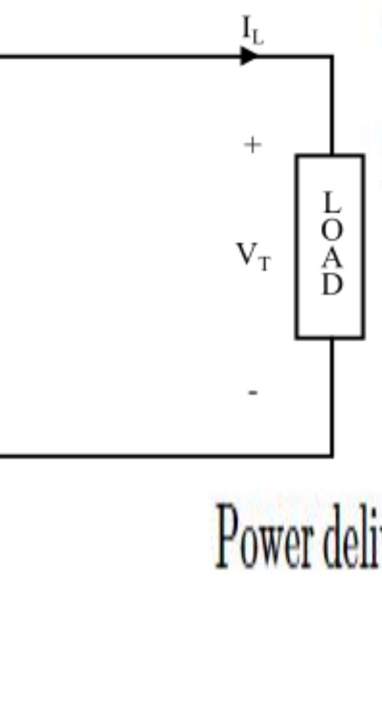
f Excited DC Generator

or

generator, the field winding is connected in series of the generator so that whole armature current h the field winding as well as the load. Since the through the field winding of the generator, so the few turns of thick wire having low resistance. The rs are used in special applications like boosters.

Excited DC Generator

ator



Armature current, $I_a = I_{se} = I_L = I(\text{Say})$

Terminal voltage, $V_T = E_g - I(R_a + R_{se})$

Power developed in armature $= E_g I_a$

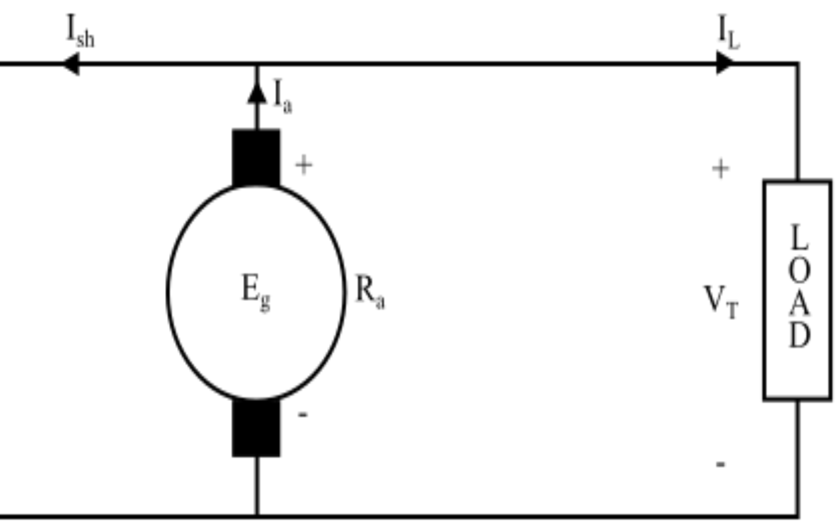
Power delivered to load $= E_g I_a - I_a^2(R_a + R_{se}) = V_T I_a = V_T I_L$

f Excited DC Generator

tor

shunt generator, the field winding is
rallel with the armature of the generator
voltage of the generator is applied across
eld winding has many turns of thin wire
stance so that only a fraction of armature
rough the shunt field winding and the rest
ne load.

xcited DC Generator



Armature current, $I_a = I_L + I_{sh}$

Shunt field current, $I_{sh} = \frac{V_T}{R_{sh}}$

erminal voltage, $V_T = E_g - I_a R_a$

wer developed in armature $= E_g I_a$

Power delivered to load $= V_T I_L$

Excited DC Generator

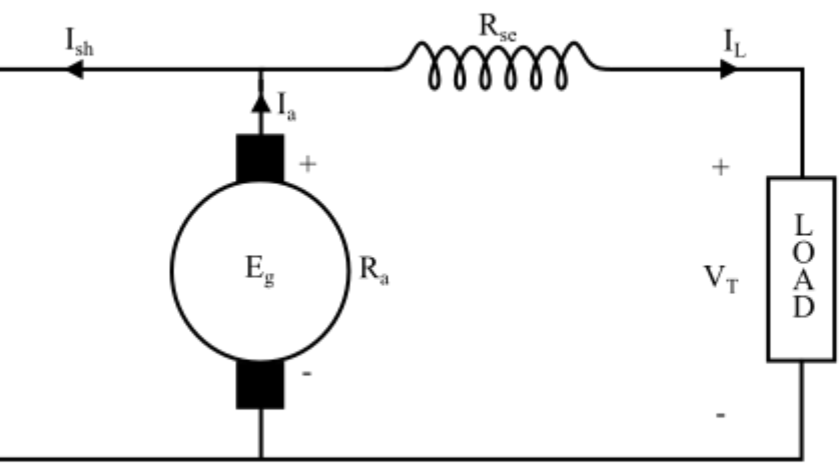
nerator

pound generator, there are two field winding one is in series and the other is in parallel re. The DC compound generators are of two

ompound Generator

nt generator, only shunt field winding is allel with the armature.

Excited DC Generator



Series field current, $I_{se} = I_L$

hunt field current, $I_{sh} = \dfrac{V_T + I_{se}R_{se}}{R_{sh}}$

minal voltage, $V_T = E_g - I_aR_a - I_{se}R_{se}$

ower developed in armature $= E_gI_a$

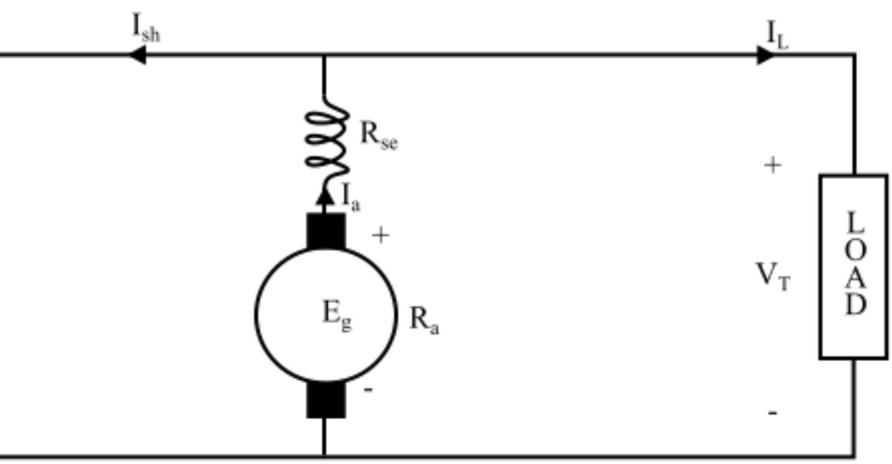
Power delivered to load $= V_TI_L$

Excited DC Generator

Compound Generator

shunt generator, the shunt field nnected in parallel with both series ature winding.

xcited DC Generator



eries field current, $I_{se} = I_a = I_L + I_{sh}$

Shunt field current, $I_{sh} = \dfrac{V_T}{R_{sh}}$

rminal voltage, $V_T = E_g - I_a(R_a + R_{se})$

Power developed in armature $= E_g I_a$

Power delivered to load $= V_T I_L$

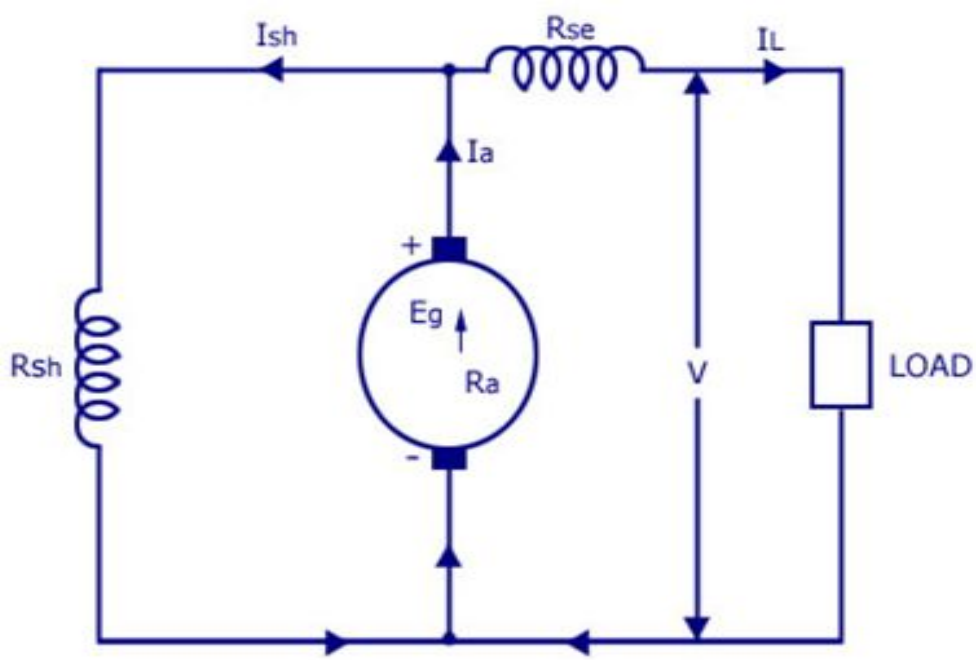
Excited DC Generator

enerator

ies field winding & shunt field winding

ompound Generator

g is parallel with armature windings



Short Shunt Compound

$I_{se} R_{se}$

$P_g = E_g I_a$

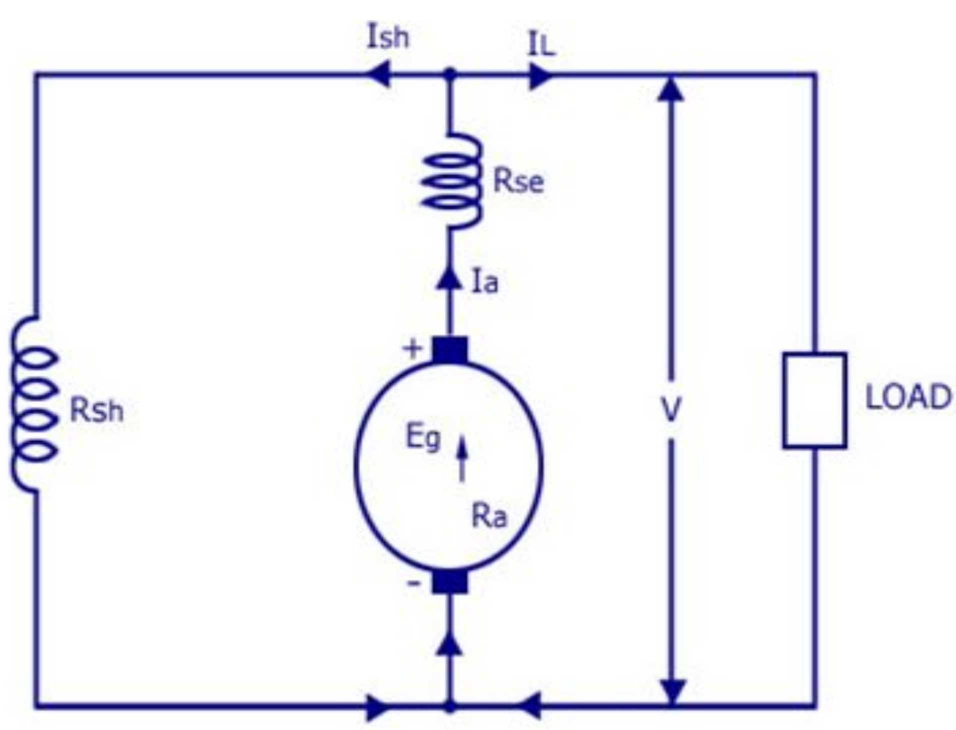
$P_o = V I_L$

Excited DC Generator

enerator

ompound Generator

is parallel with both armature and series field



Long Shunt Compound

$R_{se})$

$P_g = E_g I_a$

$P_o = VI_L$